

FAST PATH — safety-critical, deterministic, per-tick

SLOW PATH — event-triggered LLM supervision

Trustworthiness & traceability — consensus snapshots · verifier verdicts & reasons · LLM proposals · refinement cycles · safety & latency metrics

Safe control action

actuator command within
SLA / slew limits

Safe deterministic fallback

invalid / late / unsafe LLM output
→ conservative rule-based action

Consensus snapshot

state estimate $\hat{x}(t)$
confidence $\gamma(t)$, anomaly flags

Deterministic verifier

freshness · confidence · actuator
bounds · thermal SLA (NON-bypassable)

Edge moderating agents — lightweight fuzzy consensus

FPR--OWA + robust baselines (mean / trimmed / Hampel / median / direct OWA)
reliability scoring → fuzzy preference relation → quantifier-guided OWA

Sensing agents

noisy / potentially attacked thermal readings $s_{ij}(t) = x(t) + \text{noise} + \text{attack}$

Optional supervisory action

must pass the SAME deterministic
verifier before any actuation

Event trigger

fires only on low confidence /
anomaly / interval (sparse)

**LangGraph orchestrator
+ Gemma planner (LLM)**

ReAct-style reasoning over
consensus snapshots; proposes
supervisory control actions
(~7 s per call)

actuation

— Per-tick deterministic data/control flow (fast path)

— Sparse, event-triggered LLM supervision (slow path)

— Verified safe actuation